When a sound wave tries to pass from air into liquid, only a small fraction of the sound is transmitted through the interface, while the remainder of the energy is reflected. This is because air has a *low* mechanical impedance (low acoustic pressure and high particle velocity resulting from low density and high compressibility), while liquid has a *high* mechanical impedance. In less technical terms, it requires more effort to wave your hand in water than it does to wave it in air. This difference in mechanical impedance results in most of the sound being reflected at an air/liquid interface.

The middle ear is an *impedance matching* network that increases the fraction of sound energy entering the liquid of the inner ear. For example, fish do not have an ear drum or middle ear, because they have no need to hear in air. Most of the impedance conversion results from the difference in *area* between the ear drum (receiving sound from the air) and the *oval window* (transmitting sound into the liquid, see Fig. 22-1). The ear drum has an area of about 60 $(mm)^2$, while the oval window has an area of roughly 4 $(mm)^2$. Since pressure is equal to force divided by area, this difference in area increases the sound wave pressure by about 15 times.

Contained within the cochlea is the *basilar membrane*, the supporting structure for about 12,000 sensory cells forming the *cochlear nerve*. The basilar membrane is stiffest near the oval window, and becomes more flexible toward the opposite end, allowing it to act as a *frequency spectrum analyzer*. When exposed to a high frequency signal, the basilar membrane resonates where it is stiff, resulting in the excitation of nerve cells close to the oval window. Likewise, low frequency sounds excite nerve cells at the far end of the basilar membrane. This makes specific fibers in the cochlear nerve respond to specific frequencies. This organization is called the **place principle**, and is preserved throughout the auditory pathway into the brain.

Another information encoding scheme is also used in human hearing, called the **volley principle**. Nerve cells transmit information by generating brief electrical pulses called *action potentials*. A nerve cell on the basilar membrane can encode audio information by producing an action potential in response to each cycle of the vibration. For example, a 200 hertz sound wave can be represented by a neuron producing 200 action potentials per second. However, this only works at frequencies below about 500 hertz, the maximum rate that neurons can produce action potentials. The human ear overcomes this problem by allowing several nerve cells to take turns performing this single task. For example, a 3000 hertz tone might be represented by *ten* nerve cells alternately firing at 300 times per second. This extends the range of the volley principle to about 4 kHz, above which the place principle is exclusively used.

Table 22-1 shows the relationship between sound intensity and perceived loudness. It is common to express sound intensity on a logarithmic scale, called **decibel SPL** (Sound Power Level). On this scale, 0 dB SPL is a sound wave power of 10^{-16} watts/cm^2, about the weakest sound detectable by the human ear. Normal speech is at about 60 dB SPL, while painful damage to the ear occurs at about 140 dB SPL.